NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Computer Science & Engineering

MID SEMESTER EXAMINATION, Oct. 2023

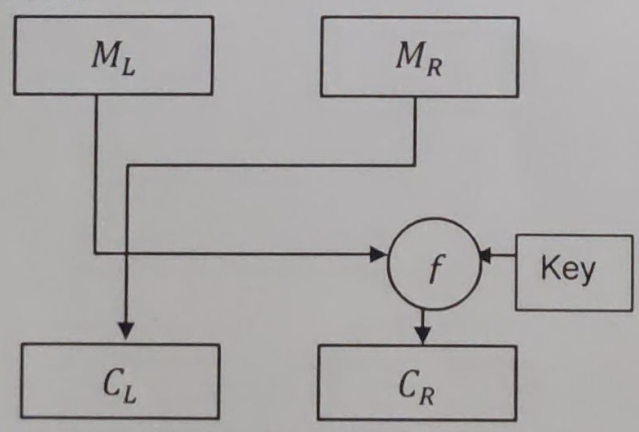
B. Tech: CSE, Semester-V

Course Name: Information Security **Code: CS54119**

Maximum Time: 2 hours **Max. Marks: 30**

Instruction:

1. ***Attempt all questions.***
2. ***Assume any suitable data, if necessary. Calculator not allowed.***
3. *The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand margin.*

		Marks	CO	BL
1.	The police department website was attacked to disrupt normal traffic of a server by overwhelming the target or its surrounding infrastructure with a barrage of traffic. Due to which the department faced a temporary problem of lost communication and using the website for its normal work. a. Identify the type of attack attempted by the attackers? Justify your answer. b. Identify which principles of information security were violated. Justify your answer. c. List down possible vulnerabilities that can cause such security glitches.	2+2+2	CO3, CO5	A, E
2.	a. Step by step discuss the DES key generation algorithm. b. A Feistel cipher is a block cipher with a particular structure. The plaintext and ciphertext are each 4 bits long and the key is 3 bits long. Show the encryption and decryption results if the original plaintext is 0111 and the key is 101.  Hint: Assume that the function $f(K, M_L)$ takes the first and third bits of the key, and XOR then with the left half of the plain text block to produce the result as a 2-bit binary pattern.	3+3	CO1, CO2	R, E
3.	**a.** Describe the RSA algorithm step by step. **b.** In RSA, given e=13 and n=221, compute the values of p, q, d . **c.** Encrypt the message "HELLO" using 00 to 25 for letters A to Z. using RSA encrypt the message using key (e, n) from the above part.	2+2+2	CO 2	P, A

4.	Part a is mandatory and answer part b or part c. a. Distinguish between a substitution cipher and a transposition cipher. b. Encipher the message "This is an exercise" to provide confidentiality using following technique (For any computation encode the letter by their position in the alphabet assuming position of a=0): First apply Additive shift cipher with key = 19 and after that on the intermediate result apply Scytale cipher with key value 6 or Double transposition with key 34521. Produce the final cipher. c. Encrypt "LASTQUESTION" using the Playfair cipher and the keyword "LEMONS".	2+4	CO 1, CO 3	R, U
5.	a. Explain the possible attacks against monoalphabetic ciphers. b. In the Diffie-Hellman protocol, two communicating parties agreed with g-7, p= 23, where one party choose the secret x=3, and another party choose secret y=5, Compute the value of the shared symmetric key?	2+4	CO 1, CO 2	R, U

****End of Questions****

Level	Bloom's Taxonomy
1	Remembering (R)
2	Understanding (U)
3	Applying (P)
4	Analyzing (A)
5	Evaluate (E)
6	Create (C)

Course Outcomes:

At the end of the course, a student should have:

CO	Outcome
1.	Identify the appropriate technologies necessary to solve concrete problems related to confidentiality (cryptographic solutions), integrity (authentication such as biometric), availability (for example, intrusion detection solutions), and privacy protection.
2.	Apply cryptography and some key encryption techniques for providing secure solutions
3.	Evaluate and communicate the human role in security systems with an emphasis on ethics, vulnerabilities and training.
4.	Develop policies and procedures to manage enterprise security risks.
5.	Determine appropriate mechanisms for identification, authentication, and protecting information systems.